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Perkins et al.

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(54) **PHOTOVOLTAIC SHINGLES WITH MULTI-MODULE POWER ELECTRONICS**

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(22) Filed: **Nov. 15, 2024**

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Related U.S. Application Data

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(51) **Int. Cl.**

H02S 40/36	(2014.01)
E04D 1/30	(2006.01)
H02S 20/25	(2014.01)

(52) **U.S. Cl.**

CPC ***H02S 40/36*** (2014.12); ***E04D 1/30*** (2013.01); ***H02S 20/25*** (2014.12); *E04D 2001/308* (2013.01)

(58) **Field of Classification Search**

CPC H02S 40/36; H02S 20/25; H02S 20/23; H02S 40/34; E04D 1/30; E04D 2001/308; Y02B 10/10; Y02E 10/50

See application file for complete search history.

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(57) **ABSTRACT**

A system includes a plurality of building integrated photovoltaic modules installed on a roof deck and arranged in an array. The plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, with the first building integrated photovoltaic module being vertically adjacent to the second building integrated photovoltaic module. Each of the building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit. The power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells. The power electronics unit is

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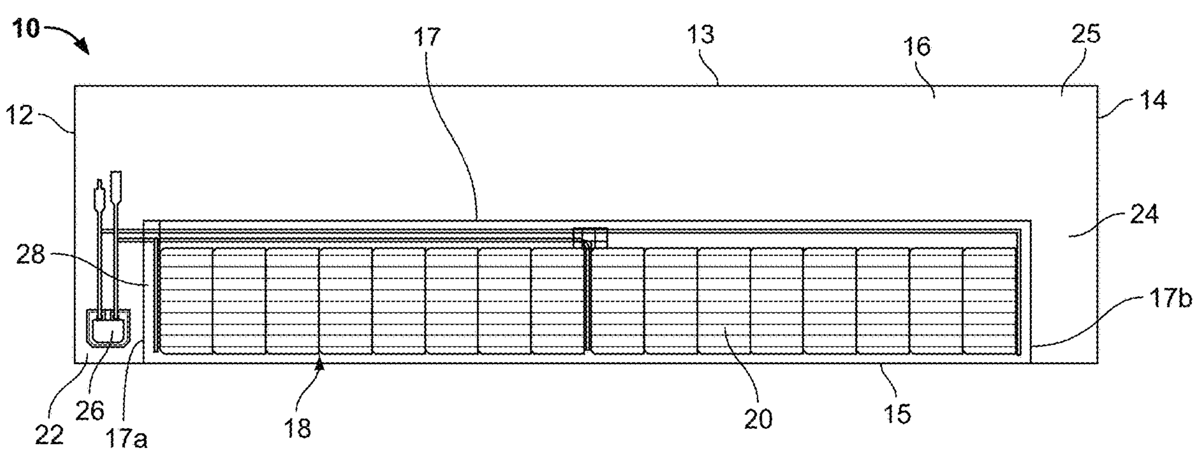

electrically connected to the plurality of solar cells. The power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module.

30 Claims, 10 Drawing Sheets

Related U.S. Application Data

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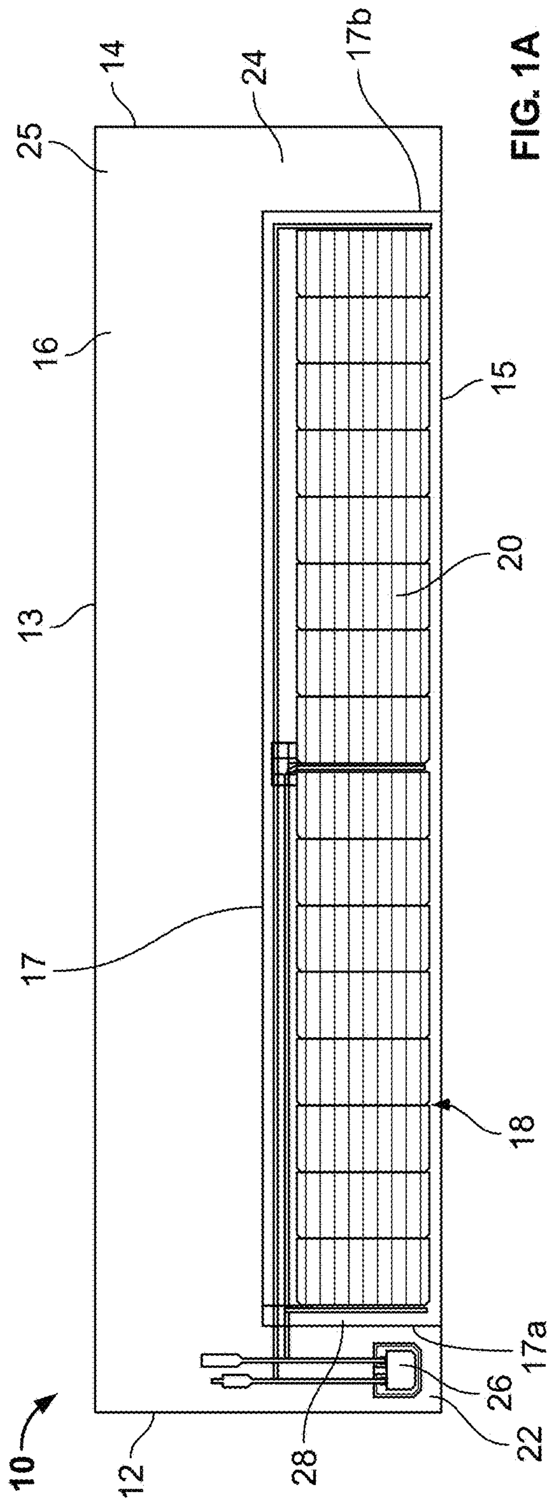


FIG. 1A

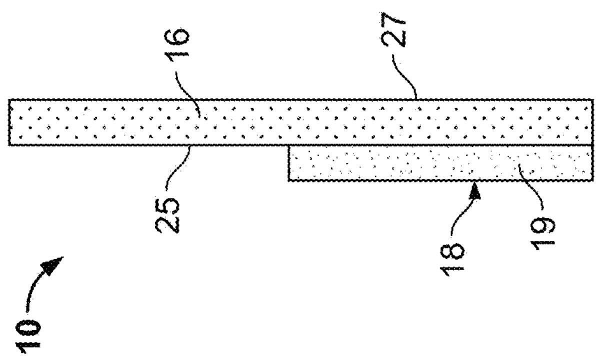
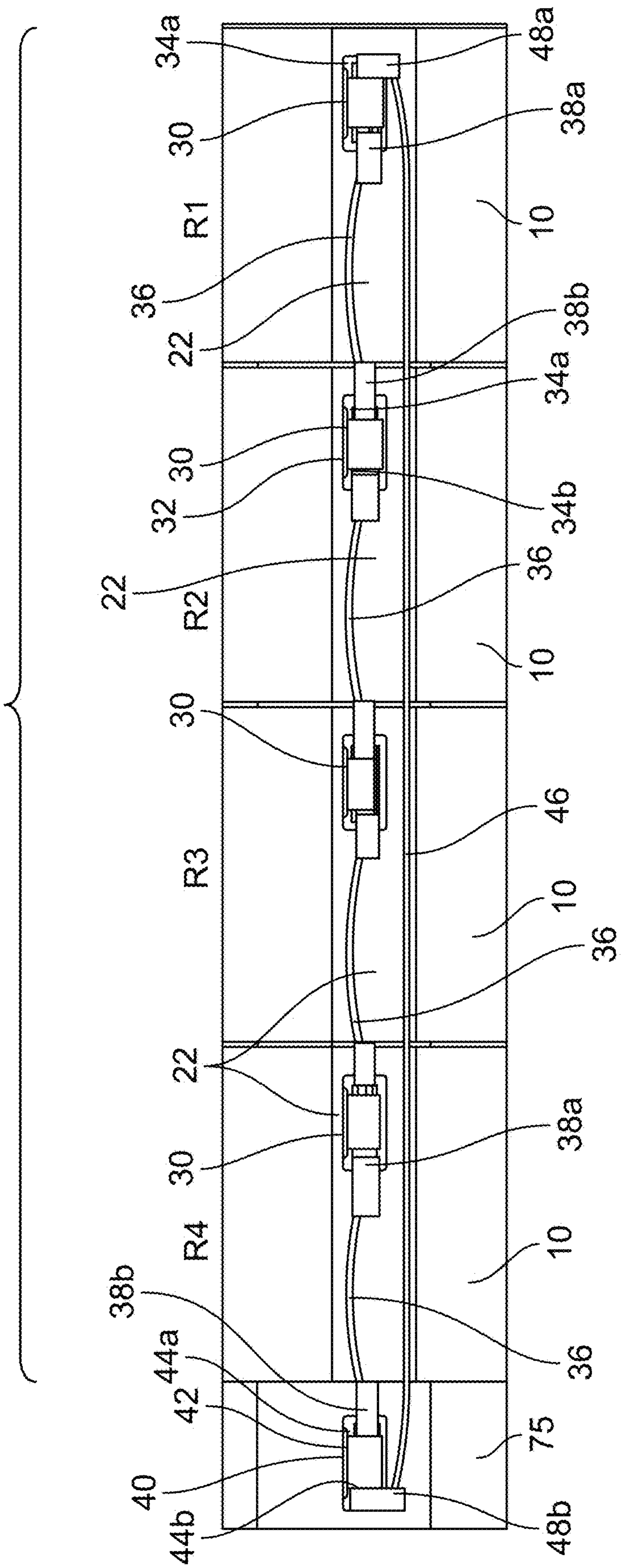


FIG. 1B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44a
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FIG. 2

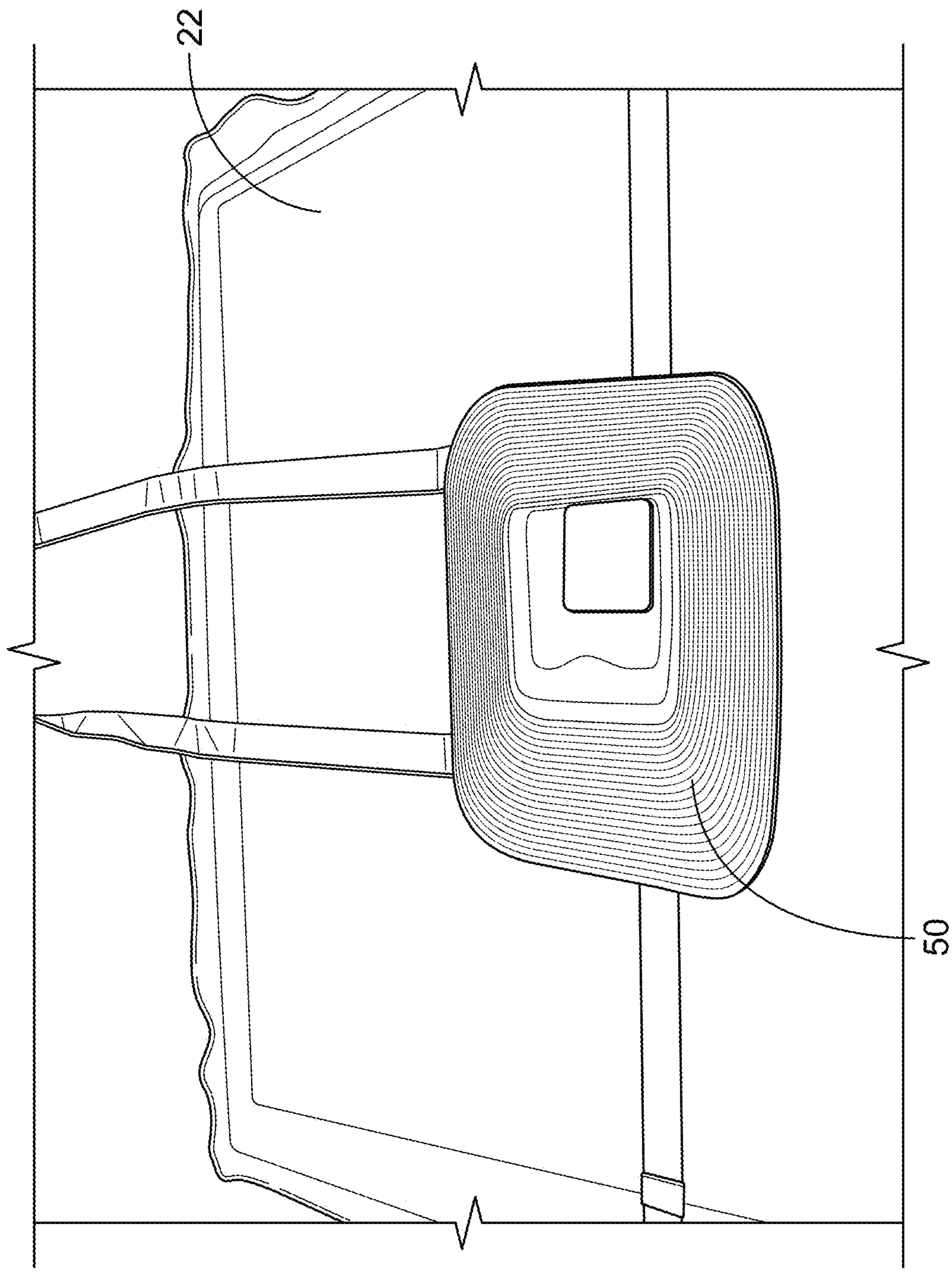
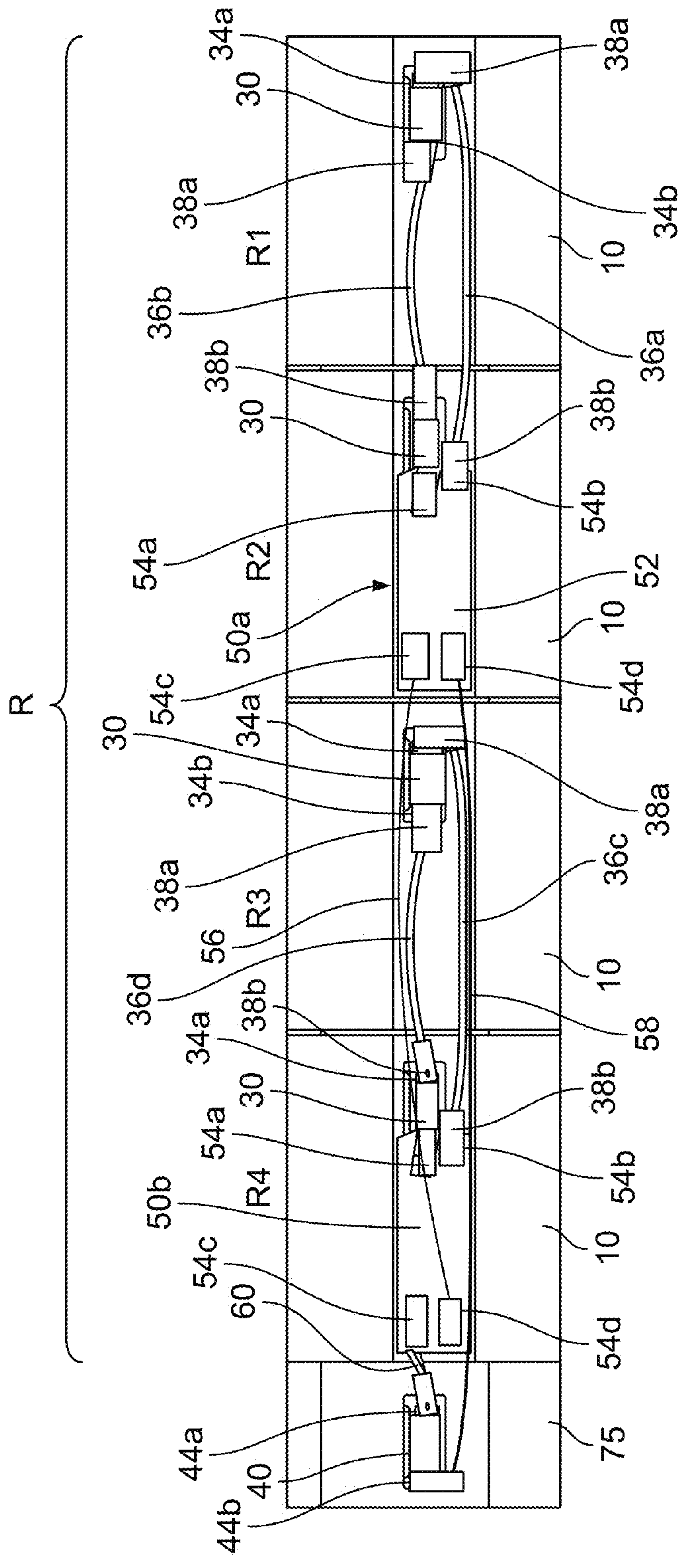


FIG. 3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R3
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54c
54d
56
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60
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FIG. 4

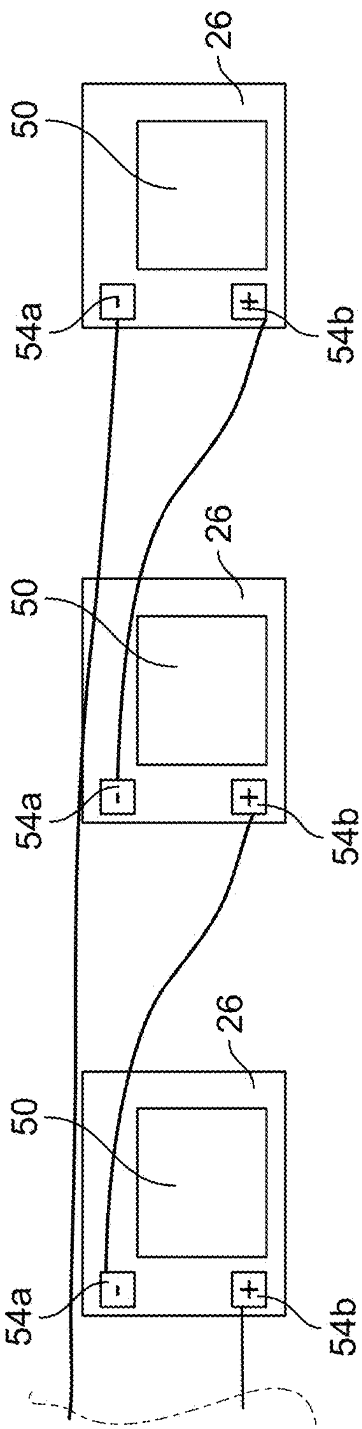

FIG. 4A

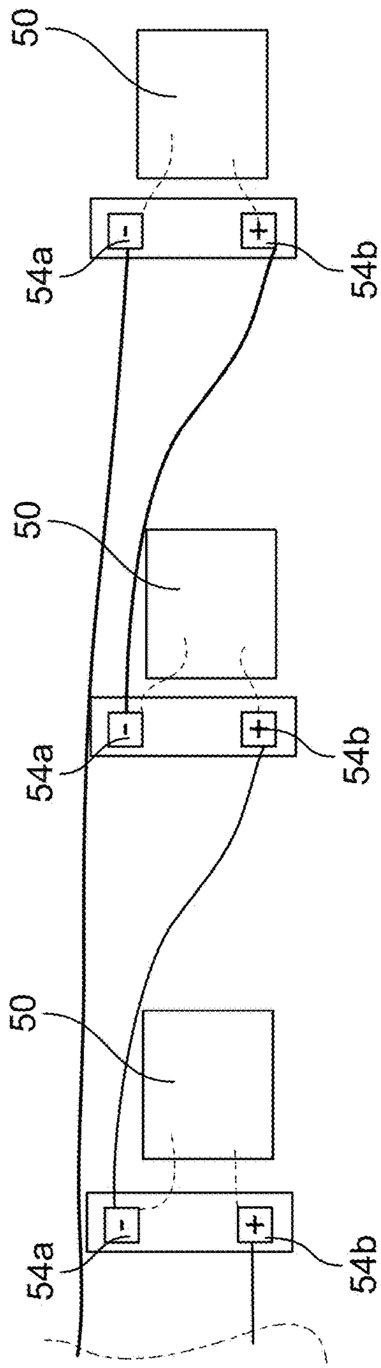

FIG. 4B

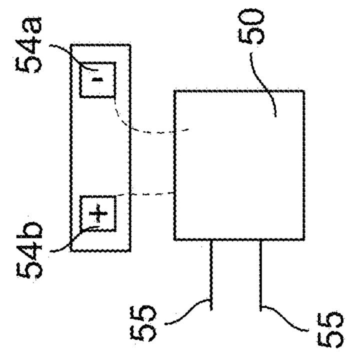

FIG. 4C

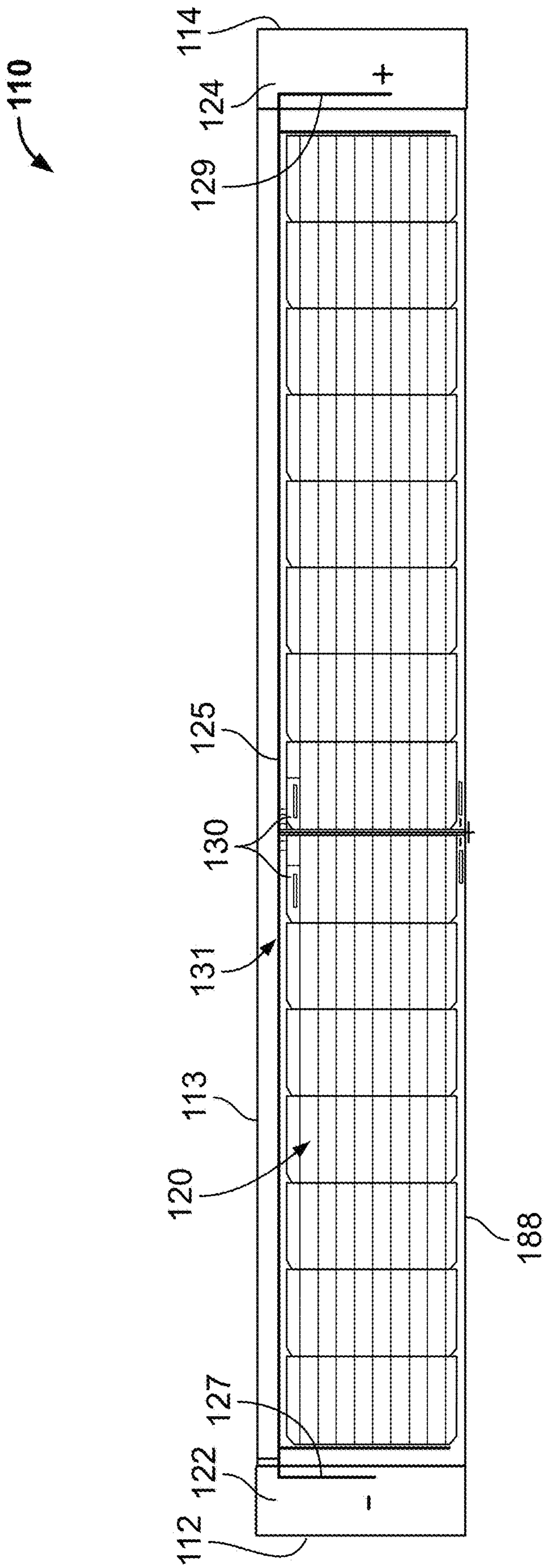

FIG. 5

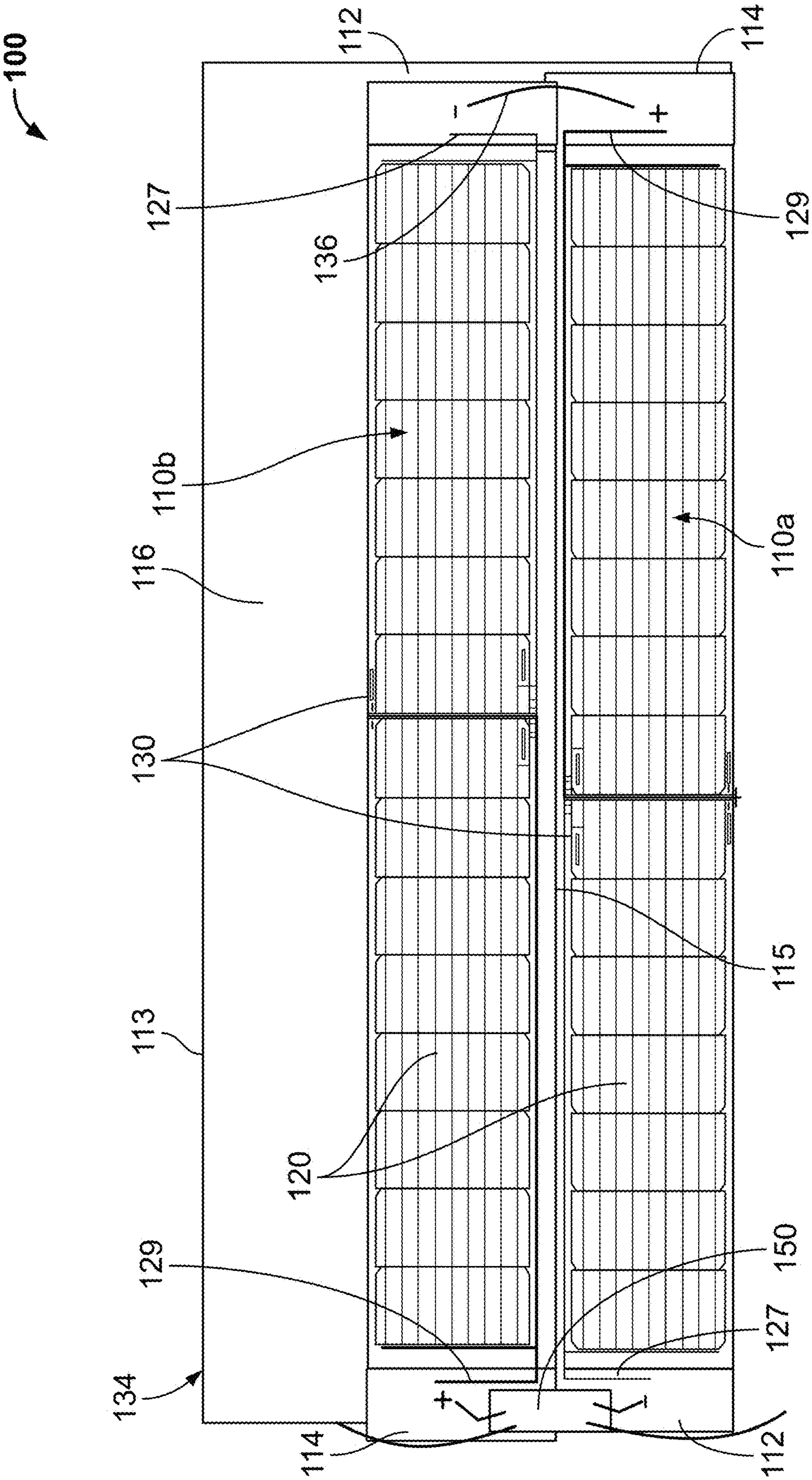
100
112
114
127
129
136
110b
110a
116
130
115
113
120
129
150
127
134
114
112
-
+
+
-

FIG. 6

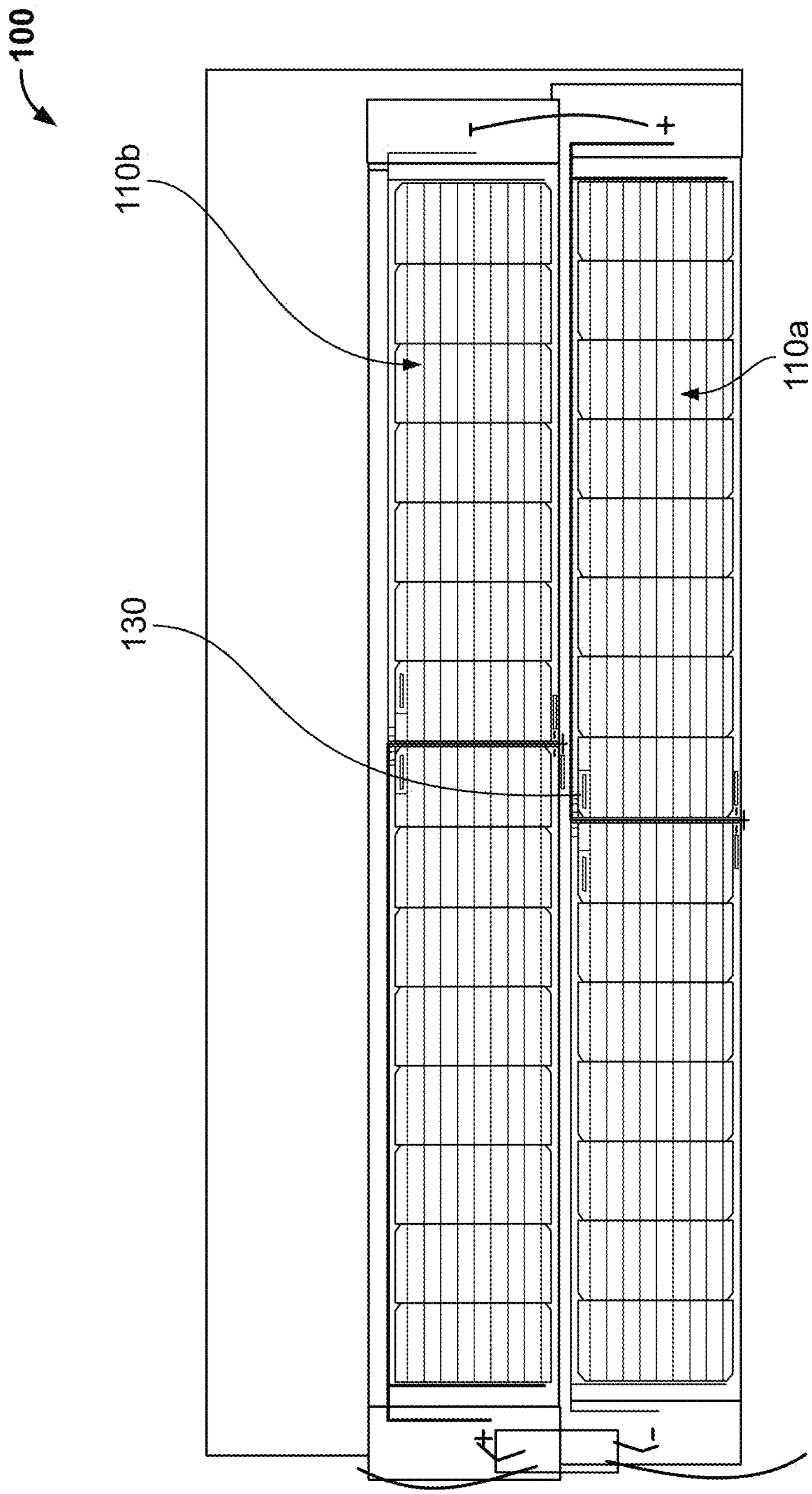

FIG. 7

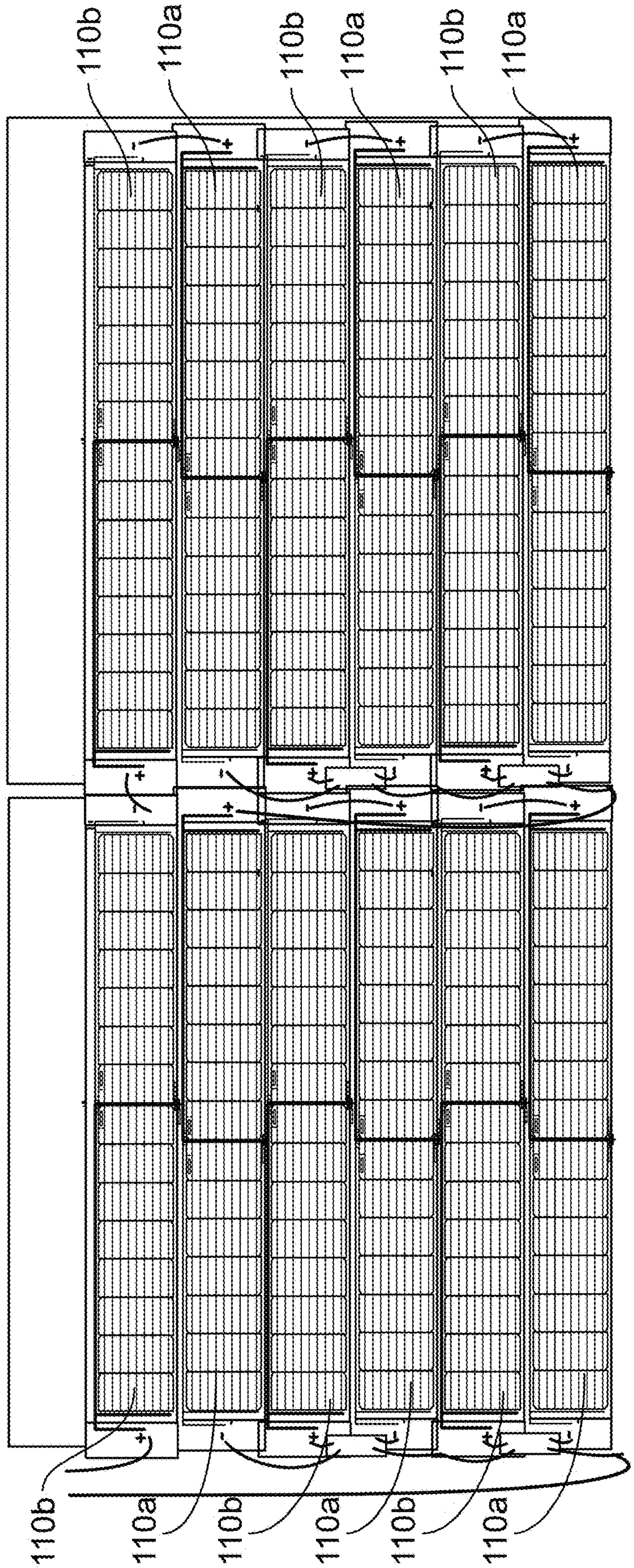

FIG. 8

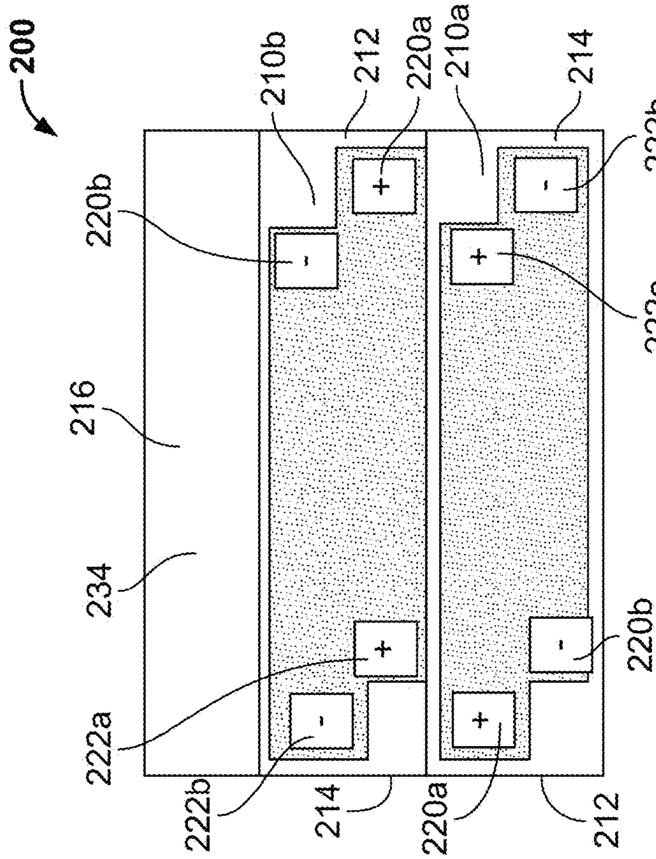

FIG. 9

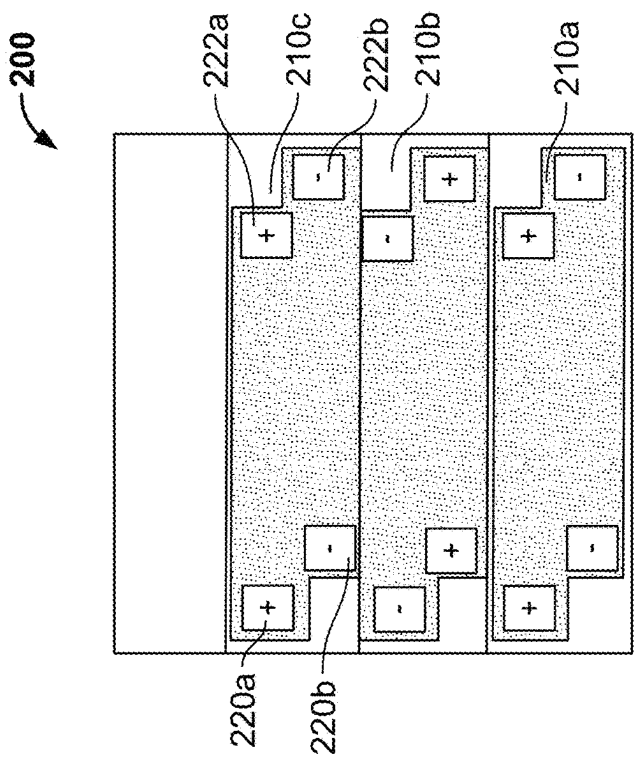

FIG. 10

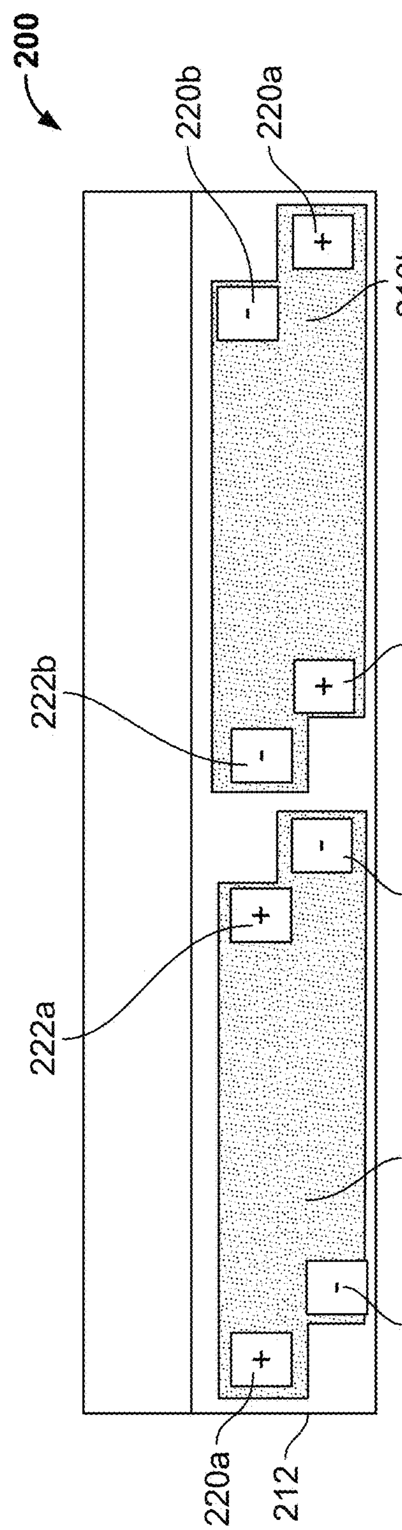

FIG. 11

PHOTOVOLTAIC SHINGLES WITH MULTI-MODULE POWER ELECTRONICS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/584,881, filed Feb. 22, 2024, which is a Section 111(a) application relating to and claiming the benefit of commonly owned, U.S. Provisional Patent Application Ser. No. 63/486,574, filed Feb. 23, 2023, entitled "PHOTOVOLTAIC SHINGLES WITH MULTI-MODULE POWER ELECTRONICS," the contents of each of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The present invention relates to photovoltaic shingles and, more particularly, photovoltaic shingles with multi-module power electronics.

BACKGROUND

Photovoltaic systems are installed on building roofs to generate electricity.

SUMMARY

In some embodiments, a system includes a plurality of building integrated photovoltaic modules installed on a roof deck, wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck, wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit, wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells, wherein the power electronics unit is electrically connected to the plurality of solar cells, and wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module.

In some embodiments, the power electronics unit is horizontally adjacent to the first end of the one of the at least one row of the plurality of solar cells. In some embodiments, the plurality of building integrated photovoltaic modules is a plurality of solar shingles. In some embodiments, the at least one row includes a plurality of rows, wherein the plurality of rows includes a first row and a second row above the first row, wherein each of the plurality of building integrated photovoltaic modules includes a first electrical connector proximate to the first end of the first row and electrically connected to the first row of the plurality of solar cells, and a second electrical connector proximate to a first end of the second row and electrically connected to the second row of the plurality of solar cells, and wherein the power electronics unit is connected to the first electrical connector and the second electrical connector.

In some embodiments, the power electronics unit includes a first terminal and a second terminal, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module. In some embodiments, each of the first electrical connector and the second electrical connector includes a first connector and a second connector, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module.

In some embodiments, the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the second electrical connector of the second building integrated photovoltaic module.

In some embodiments, the first terminal of the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module. In some embodiments, the first terminal of the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module. In some embodiments, the second terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the first building integrated photovoltaic module by a first electrical cable. In some embodiments, the second connector of the second electrical connector of the first building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the first building integrated photovoltaic module by a second electrical cable. In some embodiments, the second terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the second building integrated photovoltaic module by a third electrical cable. In some embodiments, the second connector of the second electrical connector of the second building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the second building integrated photovoltaic module by a fourth electrical cable. In some embodiments, the power electronics unit includes a third terminal and a fourth terminal, wherein the third terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the fourth terminal of the power electronics unit of the second building integrated photovoltaic module by a fifth electrical cable.

In some embodiments, the system further includes a third electrical connector installed on the roof deck, wherein each of the power electronics unit of the first building integrated photovoltaic module and the power electronics unit of the second building integrated photovoltaic module is electrically connected to the third electrical connector. In some embodiments, the third electrical connector includes a first connector and a second connector, wherein the third terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the third electrical connector by a sixth electrical cable. In some embodiments, the fourth terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the third electrical connector by a seventh electrical cable.

In some embodiments, the power electronics unit includes a housing and power electronics within the housing. In some embodiments, the power electronics includes an optimizer, a bypass diode, system monitoring electronic components, a rapid shutdown device, an electronic communication component, or combinations thereof.

In some embodiments, each of the plurality of rows includes a second end opposite the first end of the row, wherein each of the first building integrated photovoltaic module and the second building integrated photovoltaic module includes a first negative electrical terminal located at the first end of the first row thereof, a first positive electrical terminal located at the second end of the second row thereof, a second positive electrical terminal located at the first end of the second row thereof, and a second negative terminal located at the second end of the second row thereof.

In some embodiments, the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners. In some embodiments, the plurality of building integrated photovoltaic modules includes a third building integrated photovoltaic module, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module. In some embodiments, the plurality of building integrated photovoltaic modules includes a fourth building integrated photovoltaic module, wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module. In some embodiments, the fourth building integrated photovoltaic module is vertically adjacent to the third building integrated photovoltaic module. In some embodiments, the power electronics unit of the third building integrated photovoltaic module is electrically connected to the power electronics unit of the fourth building integrated photovoltaic module.

In some embodiments, a system includes a plurality of building integrated photovoltaic modules installed on a roof deck, wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck, wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row; a first power electronics unit, wherein the first power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the first building integrated photovoltaic module, wherein the first power electronics unit is electrically connected to the plurality of solar cells of the first building integrated photovoltaic module; and a second power electronics unit, wherein the second power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the second building integrated photovoltaic module, and wherein the first power electronics unit is electrically connected to the second power electronics unit.

In some embodiments, a method includes obtaining a plurality of building integrated photovoltaic modules, wherein each of the plurality of building integrated photovoltaic modules includes a plurality of solar cells arranged in at least one row, and a power electronics unit, wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells, wherein the power electronics unit is electrically connected to the plurality of solar cells; and installing at least a first building integrated photovoltaic module and a second building integrated photovoltaic module on a roof deck, wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, and wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module.

In some embodiments, the method further includes installing a third building integrated photovoltaic module and a fourth building integrated photovoltaic module on the roof deck, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module, and wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module and vertically adjacent to the third building integrated photovoltaic module. In some embodiments, the plurality of building integrated photovoltaic modules is a plurality of solar shingles. In some embodiments, the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. **1**A and **1**B are a top plan view and a side elevational view, respectively, of some embodiments of a photovoltaic shingle;

FIG. **2** is a top plan view of some embodiments of a portion of a roofing system including photovoltaic shingles;

FIG. **3** illustrates some embodiments of a power electronics unit;

FIG. **4** is a top plan view of some embodiments of a portion of a roofing system including photovoltaic shingles and power electronics units;

FIGS. **4**A through **4**C are schematic views of some embodiments of power electronics for photovoltaic shingles;

FIG. **5** is a top plan view of some embodiments of a photovoltaic module;

FIG. **6** is a top plan view of some embodiments of a photovoltaic shingle including photovoltaic modules shown in FIG. **5**;

FIG. **7** is a top plan view of some embodiments of a photovoltaic module;

FIG. **8** is a top plan view of some embodiments of a roofing system including photovoltaic shingles having the photovoltaic modules shown in FIG. **7**; and

FIGS. **9** through **11** are schematic, top plan views of some embodiments of photovoltaic shingles.

DETAILED DESCRIPTION

Referring to FIGS. **1**A and **1**B, in some embodiments, a photovoltaic shingle **10** includes a first end **12**, a second end **14** opposite the first end **12**, a first edge **13** extending from the first end **12** to the second end **14**, and a second edge **15** opposite the first edge **13** and extending from the first end **12** to the second end **14**. In some embodiments, the photovoltaic shingle **10** includes a headlap portion **16**. In some embodiments, the headlap portion **16** extends from the first end **12** to the second end **14** and from the first edge **13** to a first location **17** between the first edge **13** and the second edge **15**. In some embodiments, the photovoltaic shingle **10** includes a reveal portion **18**. In some embodiments, the reveal portion **18** includes a photovoltaic layer **19**. In some embodiments, the photovoltaic layer **19** includes at least one solar cell **20**. In some embodiments, the photovoltaic shingle **10** includes a first side lap portion **22** located at the first end **12**. In some embodiments, the first side lap portion **22** includes a width extending from the first end **12** to a second location **17*a*** between the first end **12** and the second end **14**. In some embodiments, the first side lap portion **22** includes a length extending from the first location **17** to the second edge **15**. In some embodiments, the photovoltaic shingle **10** includes a second side lap portion **24** located at the second end **14**. In some embodiments, the second side lap portion **24** includes a width extending from the second end **14** to a third location **17*b*** between the first end **12** and the second end **14**. In some embodiments, the second side lap portion **24** includes a length extending from the first location **17** to the second edge **15**. In some embodiments, the photovoltaic shingle **10** includes an outer surface **25** and an inner surface **27** opposite the outer surface **25**. In some embodiments, the reveal portion **18** extends from the first side lap portion **22** to the second side lap portion **24** and from the second edge **15** to the first location **17**. In some embodiments, the photovoltaic shingle **10** is configured to be installed on a building structure. In some embodiments, the photovoltaic shingle **10** is configured to be installed on a roof deck. In some embodiments, at least one junction box **26** is located on the first side lap portion **22**. In some embodiments, the at least one junction box **26** includes a plurality of the junction boxes **26**. In certain embodiments, other electronic and electrical components may be attached to the first side lap portion **22**. In some embodiments, non-limiting examples of such electronic and electrical components include an electrical connector, a rapid shutdown device, an optimizer, and an inverter. In some embodiments, the photovoltaic shingle **10** includes a structure, composition, components, and/or function similar to those of one or more embodiments of the photovoltaic shingles disclosed in in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled "Building Integrated Photovoltaic System," owned by GAF Energy LLC, and U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled "Photovoltaic Shingles and Methods of Installing Same," the contents of each of which are incorporated by reference herein in their entirety.

In some embodiments, the at least one solar cell **20** includes a plurality of the solar cells **20**. In some embodiments, the plurality of solar cells **20** includes two solar cells. In some embodiments, the plurality of solar cells **20** includes three solar cells. In some embodiments, the plurality of solar cells **20** includes four solar cells. In some embodiments, the plurality of solar cells **20** includes five solar cells. In some embodiments, the plurality of solar cells **20** includes six solar cells. In some embodiments, the plurality of solar cells **20** includes seven solar cells. In some embodiments, the plurality of solar cells **20** includes eight solar cells. In some embodiments, the plurality of solar cells **20** includes nine solar cells. In some embodiments, the plurality of solar cells **20** includes ten solar cells. In some embodiments, the plurality of solar cells **20** includes eleven solar cells. In some embodiments, the plurality of solar cells **20** includes twelve solar cells.

In some embodiments, the plurality of solar cells **20** includes thirteen solar cells. In some embodiments, the plurality of solar cells **20** includes fourteen solar cells. In some embodiments, the plurality of solar cells **20** includes fifteen solar cells. In some embodiments, the plurality of solar cells **20** includes sixteen solar cells. In some embodiments, the plurality of solar cells **20** includes more than sixteen solar cells.

In some embodiments, the plurality of solar cells **20** is arranged in one row (i.e., one reveal). In another embodiment, the plurality of solar cells **20** is arranged in two rows (i.e., two reveals). In another embodiment, the plurality of solar cells **20** is arranged in three rows (i.e., three reveals). In another embodiment, the plurality of solar cells **20** is arranged in four rows (i.e., four reveals). In another embodiment, the plurality of solar cells **20** is arranged in five rows (i.e., five reveals). In another embodiment, the plurality of solar cells **20** is arranged in six rows (i.e., six reveals). In other embodiments, the plurality of solar cells **20** is arranged in more than six rows. In some embodiments, each of the rows of the plurality of solar cells **20** includes a first end **28** located proximate to the first side lap portion **22**.

In some embodiments, a roofing system includes a plurality of the photovoltaic shingles **10** installed on a roof deck. In some embodiments, the roof deck is a steep slope roof deck. As defined herein, a "steep slope roof deck" is any roof deck that is disposed on a roof having a pitch of Y/X, where Y and X are in a ratio of 4:12 to 12:18, where Y corresponds to the "rise" of the roof, and where X corresponds to the "run" of the roof.

In some embodiments, the plurality of photovoltaic shingles **10** is installed directly to the roof deck. In some embodiments, each of the plurality of photovoltaic shingles **10** is installed on the roof deck by a plurality of fasteners. In some embodiments, the plurality of fasteners is installed through the headlap portion **16**. In some embodiments, the plurality of fasteners includes a plurality of nails. In some embodiments, the plurality of fasteners includes a plurality of rivets. In some embodiments, the plurality of fasteners includes a plurality of screws. In some embodiments, the plurality of fasteners includes a plurality of staples.

In some embodiments, each of the plurality of photovoltaic shingles **10** is installed on the roof deck by an adhesive. In some embodiments, the adhesive is adhered directly to the roof deck. In some embodiments, the adhesive is adhered to an underlayment. In some embodiments, the underlayment is adhered directly to the roof deck. In some embodiments, the adhesive is located on a rear surface of the photovoltaic shingle **10**. In some embodiments, the adhesive includes at least one adhesive strip. In some embodiments, the adhesive includes a plurality of adhesive strips. In some embodiments, the plurality of adhesive strips is arranged intermittently. In some embodiments, the adhesive is located proximate to one edge of the photovoltaic shingle **10**. In some embodiments, the adhesive is a peel and stick film sheet. In some embodiments, the peel and stick film sheet includes at least one sheet of film removably attached to the rear surface. In some embodiments, the peel and stick film sheet is composed of EverGuard Freedom HW peel and stick membrane manufactured by GAF. In some embodiments, the adhesive includes polyvinyl butyrate, acrylic, silicone, or polycarbonate. In some embodiments, the adhesive includes pressure sensitive adhesives.

In some embodiments, a first photovoltaic shingle **10** is vertically adjacent to a second photovoltaic shingle **10**. In some embodiments, the first photovoltaic shingle **10** overlays at least a part of the headlap portion **16** of the second photovoltaic shingle **10**.

Referring to FIG. **2**, in some embodiments, a roofing system includes a plurality of photovoltaic shingles **10** installed on a roof deck **75** in a plurality of rows R. In some embodiments, each of a plurality of the photovoltaic shingles **10** includes a first electrical connector **30**. In some embodiments, the first electrical connector **30** is located on the first side lap portion **22**. In some embodiments, the first electrical connector **30** is located on the first side lap portion **22** proximate to the first end **28** of the row of solar cells **20**. In some embodiments, the first electrical connector **30** is at least partially embedded within the side lap portion **22**. In some embodiments, the first electrical connector **30** is attached to an upper surface of the side lap portion **22**. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by an adhesive. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by welding. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by heat welding. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by ultrasonic welding. In some embodiments, the first electrical connector **30** is attached to the upper surface of the side lap portion **22** by at least one fastener. In some embodiments, the first electrical connector **30** is removably attached to the upper surface of the side lap portion **22**.

In some embodiments, the first electrical connector **30** includes a structure, composition, components, and/or function similar to those of one or more embodiments of the electrical connectors disclosed in U.S. Patent Application Publication No. 2022/0029037 to Nguyen et al., published Jan. 27, 2022 and entitled "Photovoltaic Systems" and U.S. Patent Application Publication No. 2022/0311377 to Bunea et al., published Sep. 29, 2022 and entitled "Electrical Components for Photovoltaic Systems," the contents of each of which are incorporated by reference herein in its entirety.

In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 6 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 5 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 4 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 3 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm to 2 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 6 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 5 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 4 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm to 3 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 6 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 5 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm to 4 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm to 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm to 6 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm to 5 mm. In some embodiments, the first electrical connector **30** has a thickness of 5 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 5 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 5 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 5 mm to 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 5 mm to 6 mm.

In some embodiments, the first electrical connector **30** has a thickness of 6 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 6 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 6 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 6 mm to 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 7 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 7 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 7 mm to 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 8 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 8 mm to 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 9 mm to 10 mm. In some embodiments, the first electrical connector **30** has a thickness of 1 mm. In some embodiments, the first electrical connector **30** has a thickness of 2 mm. In some embodiments, the first electrical connector **30** has a thickness of 3 mm. In some embodiments, the first electrical connector **30** has a thickness of 4 mm. In some embodiments, the first electrical connector **30** has a thickness of 5 mm. In some embodiments, the first electrical connector **30** has a thickness of 6 mm. In some embodiments, the first electrical connector **30** has a thickness of 7 mm. In some embodiments, the first electrical connector **30** has a thickness of 8 mm. In some embodiments, the first electrical connector **30** has a thickness of 9 mm. In some embodiments, the first electrical connector **30** has a thickness of 10 mm.

In some embodiments, the first electrical connector **30** is electrically connected to the plurality of solar cells **20**. In some embodiments, the first electrical connector **30** is electrically connected to the plurality of solar cells **20** by electrical bussing. In some embodiments, the electrical bussing is embedded within the photovoltaic shingle **10**.

In some embodiments, the first electrical connector **30** includes a housing **32** and a pair of connectors **34*a***, **34*b***. In some embodiments, the connectors **34*a***, **34*b*** extend from the housing **32**. In some embodiments, the connectors **34***a*, **34***b* are snap-fit connectors. In some embodiments, the connector **34***a* is spaced apart from the connector **34***b*. In some embodiments, the connector **34***a* is located proximate to one end of the housing **32** and the connector **34***b* is located proximate to an opposite end of the housing **32**. In some embodiments, each of the first electrical connectors **30** is electrically connected to one another by a plurality of electrical jumper wires **36** (also referred to as jumper wires, electrical jumper cables, or jumper cables). In some embodiments, the first electrical connectors **30** are electrically connected to one another in series by the plurality of jumper wires **36**. In some embodiments, each of the jumper wires **36** includes a first connector **38***a* at a first end thereof and a second connector **38***b* at a second, opposite end thereof. In some embodiments, the first connector **38***a* of a corresponding one of the jumper wires **36** is connected to a corresponding one of the connector **34***b* of the housing **32** of the first electrical connector **30**. In some embodiments, the first connector **38***a* of a corresponding one of the jumper wires **36** is preconnected to a corresponding one of the connector **34***b* of the housing **32** of the first electrical connector **30** of the photovoltaic shingle **10** prior to installation on the roof deck **75**. In some embodiments, the second connector **38***b* of a corresponding one of the jumper wires **36** is connected to a corresponding one of the connector **34***a* of the housing **32** of the first electrical connector **30** of an adjacent lower one of the plurality of photovoltaic shingles **10** in rows R**2**, R**3**, R**4**.

In some embodiments, the system includes a second electrical connector **40**. In some embodiments, the second electrical connector **40** is installed on the roof deck **75**. In some embodiments, the second electrical connector **40** is located below row R**4**. In some embodiments, the second electrical connector **40** is proximate to row R**4**. In some embodiments, the second electrical connector **40** is adjacent row R**4**. In some embodiments, the second electrical connector **40** includes a housing **42** and a pair of connectors **44***a*, **44***b*. In some embodiments, the connectors **44***a*, **44***b* extend from the housing **42**. In some embodiments, the second electrical connector **40** has a similar structure and/or function of those of the first electrical connector **30**. In some embodiments, a return wire **46** electrically connects the first electrical connector **30** of the photovoltaic shingle **10** in the uppermost row R**1** with the second electrical connector **40**. In some embodiments, the return wire **46** includes a first connector **48***a* at a first end thereof and a second connector **48***b* at a second, opposite end thereof. In some embodiments, the connector **48***a* of the return wire **46** is connected to the connector **34***a* of the first electrical connector **30** of the photovoltaic shingle **10** in row R**1**. In some embodiments, the connector **48***b* of the return wire **46** is connected to the connector **44***b* of the second electrical connector **40**. In some embodiments, the return wire **46** has a length. In some embodiments, the length of the return wire **46** may vary and be selected by an installer of the system, depending upon the number of rows R of the photovoltaic shingles **10** in the column of the array thereof.

Referring to FIG. **3**, in some embodiments, the photovoltaic shingle **10** includes a power electronics unit **50**. In some embodiments, the power electronics unit **50** is located on the first side lap portion **22**. In some embodiments, the power electronics unit **50** is attached to the first side lap portion **22**. In some embodiments, the power electronics unit **50** is laminated with the first side lap portion **22**. In some embodiments, the power electronics unit **50** is a DC optimizer PCB. In some embodiments, the power electronics unit **50** may be other PCB's having other functionality, such as bypass diodes, system monitoring electronic components, electronic communication components, rapid shutdown devices, or electronic circuitry assemblies performing one or more functionalities such as power optimization, rapid-shutdown, monitoring or communication.

Referring to FIG. **4**, in some embodiments, the roofing system illustrated in FIG. **3** and described above may be configured to incorporate a plurality of the power electronics units **50**. In some embodiments, the power electronics units **50** are installed once every two rows of the photovoltaic shingles **10**. In some embodiments, the power electronics units **50** are installed in every row of the photovoltaic shingles **10**. In some embodiments, the power electronics units **50** are installed in some of the rows of the photovoltaic shingles **10**. In some embodiments, the power electronics unit **50** includes a housing **52**. In some embodiments, the housing **52** is composed of plastic. In some embodiments, the housing is composed of metal. In some embodiments, power electronics are sealed within the housing **52**. In some embodiments, the power electronics unit **50** includes four terminals **54***a*, **54***b*, **54***c*, **54***d*. In some embodiments, each of the terminals **54***a-d* is a bulkhead terminal. In some embodiments, the power electronics unit **50** may include less than four of the terminals **54***a-d*. In some embodiments, the power electronics unit **50** may include more than four of the terminals **54***a-d*.

In some embodiments, a first power electronics unit **50***a* is installed on the photovoltaic shingle **10** in row R**2**. In some embodiments, the first power electronics unit **50***a* is connected to the first electrical connector **30** of the photovoltaic shingle **10** in row R**2**. In some embodiments, the terminal **54***a* of the first power electronics unit **50***a* is connected to connector **34***a* of the first electrical connector **30** of the photovoltaic shingle **10** in row R**2**. In some embodiments, the second connector **38***b* of jumper wire **36***a* is connected to the terminal **54***b* of the first power electronics unit **50***a*, while the first connector **38***a* of the jumper wire **36***a* is connected to the connector **34***a* of the first electrical connector **30** of the photovoltaic shingle **10** in row R**1**. In some embodiments, the second connector **38***b* of jumper wire **36***b* is connected to the first connector **34***a* of the connector **30** in row R**2**, while the first connector **38***a* of the jumper wire **36***b* is connected to the second connector **34***b* of the connector **30** in row R**1**.

In some embodiments, a second power electronics unit **50***b* is installed on the photovoltaic shingle **10** in row R**4**. In some embodiments, the second power electronics units **50***b* is connected to the first electrical connector **30** of the photovoltaic shingle **10** in row R**4**. In some embodiments, the terminal **54***a* of the second power electronics unit **50***b* is connected to connector **34***a* of the first electrical connector **30** of the photovoltaic shingle **10** in row R**4**. In some embodiments, the second connector **38***b* of jumper wire **36***c* is connected to the terminal **54***b* of the second power electronics unit **50***b*, while the first connector **38***a* of the jumper wire **36***c* is connected to the connector **34***a* of the first electrical connector **30** of the photovoltaic shingle **10** in row R**3**. In some embodiments, the second connector **38***b* of jumper wire **36***d* is connected to the first connector **34***a* of the connector **30** in row R**4**, while the first connector **38***a* of the jumper wire **36***d* is connected to the second connector **34***b* of the connector **30** in row R**3**.

In some embodiments, the terminal **54***c* of the first power electronics unit **50***a* is electrically connected to terminal **54***d* of the second power electronics unit **50***b*. In some embodiments, the terminal **54***c* of the first power electronics unit **50***a* is electrically connected to terminal **54***d* of the second power electronics unit **50***b* by a jumper wire **56**. In some embodiments, the terminal **54***d* of the first power electronics unit **50***a* is electrically connected to the connector **44***b* of the second electrical connector **40**. In some embodiments, the terminal **54***d* of the first power electronics unit **50***a* is electrically connected to the connector **44***b* of the second electrical connector **40** by a jumper wire **58**. In some embodiments, the terminal **54***c* of the second power electronics unit **50***b* is electrically connected to the connector **44***a* of the second electrical connector **40**. In some embodiments, the terminal **54***c* of the second power electronics unit **50***b* is electrically connected to the connector **44***a* of the second electrical connector **40** by a jumper wire **60**. In some embodiments, the first power electronics unit **50***a* is electrically connected to the second power electronics unit **50***b*. In some embodiments, it should be understood that the roofing system may have more or less than four of the rows R1-R4 of the photovoltaic shingles **10**, and, therefore, the configuration of the power electronics units **50***a-b*, the electrical connectors **30**, **40** and wires **36***a-d*, **56**, **58**, **60** would be modified accordingly.

In some embodiments, any or all of the power electronics units **50***a-b*, the electrical connectors **30**, **40** and wires **36***a-d*, **56**, **58**, **60** may be installed by an installer on the roof deck **75**. In some embodiments, any or all of the power electronics units **50***a-b*, the electrical connectors **30**, **40** and wires **36***a-d*, **56**, **58**, **60** may be prefabricated prior to installation by an installer on the roof deck **75**. In some embodiments, at least two of the photovoltaic shingles **10** are mechanically attached to one another and to a common headlap portion using adhesives, bonding or thermal welding processes. In some embodiments, the power electronics units **50***a-b* and applicable ones of the wires **36***a-d*, **56**, **58**, **60** are attached to the photovoltaic shingles **10** that are on the common headlap portion.

Referring to FIG. **4**A, in some embodiments, each of the power electronics units **50** is located within a corresponding one of the junction boxes **26**. In some embodiments, the terminals **54***a*, **54***b* are located on or in the junction box **26**. In some embodiments, at least one of the power electronics units **50** is located within a corresponding one of the junction boxes **26**. Referring to FIG. **4**B, in some embodiments, each of the power electronics units **50** is separate from the terminals **54***a*, **54***b*. Referring to FIG. **4**C, in some embodiments, the power electronics unit **50** includes one or more electrical connections **55** to the photovoltaic module **10**, the solar cells **20**, circuit formation, or other circuitry in between.

Referring to FIG. **5**, in some embodiments, a photovoltaic module **110** includes a first end **112**, a second end **114** opposite the first end **112**, a reveal portion **118** having a plurality of solar cells **120**, a first side lap portion **122** at the first end **112**, and a second side lap portion **124** at the second end **114**. In some embodiments, the photovoltaic module **110** includes a busbar **125** extending from the first side lap portion **122** to the second side lap portion **124**. In some embodiments, the busbar **125** includes a first branch **127** located at the first side lap portion **122**. In some embodiments, the busbar **125** includes a second branch **129** located at the second side lap portion **124**. In some embodiments, the first branch **127** includes a negative electrical terminal. In some embodiments, the second branch **129** includes a positive electrical terminal. In some embodiments, the first branch **127** includes a positive electrical terminal. In some embodiments, the second branch **129** includes a negative electrical terminal. In some embodiments, the photovoltaic module **110** includes at least one bypass diode **130** electrically connected to the busbar **125**. In some embodiments, the at least one bypass diode **130** includes a plurality of bypass diodes **130**. In some embodiments, the at least one bypass diode **130** is located within a section **131** of the photovoltaic module **110** that is located between a first edge **113** of the photovoltaic module **110** and the plurality of solar cells **120**.

Referring to FIG. **6**, in some embodiments, a photovoltaic shingle **100** includes a plurality of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** is attached to a backsheet **134**. In some embodiments, the plurality of photovoltaic modules **110** overlay the backsheet **134**. In some embodiments each of the plurality of photovoltaic modules **110** is vertically stacked relative to one another on the backsheet **134**. In some embodiments, the photovoltaic shingle **100** includes a headlap portion **116**. In some embodiments, a portion of the backsheet **134** forms the headlap portion **116**. In some embodiments, the backsheet **134** includes at least one layer. In some embodiments, the backsheet **134** includes at least two layers. In some embodiments, the backsheet **134** is composed of a polymeric material. In some embodiments, In some embodiments, the backsheet **134** is composed of polyethylene terephthalate ("PET"). In some embodiments, the backsheet **134** is composed of ethylene tetrafluoroethylene ("ETFE"). In some embodiments, the backsheet **134** is composed of an acrylic such as polymethyl methacrylate ("PMMA"). In some embodiments, the backsheet **134** is composed of thermoplastic polyolefin (TPO). In some embodiments, the backsheet **134** is composed of a single ply TPO roofing membrane. In some embodiments, non-limiting examples of TPO membranes are disclosed in U.S. Pat. No. 9,359,014 to Yang et al., which is incorporated by reference herein in its entirety. In some embodiments, the backsheet **134** is composed of polyvinyl chloride. In some embodiments, the backsheet **134** is composed of ethylene propylene diene monomer (EPDM) rubber. In some embodiments, the backsheet **134** includes a flame retardant additive. In some embodiments, the flame retardant additive may be clays, nanoclays, silicas, carbon black, metal hydroxides such as aluminum hydroxide, metal foils, graphite, and combinations thereof.

In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134**. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by an adhesive. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by thermal bonding. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by ultrasonic welding. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by thermal welding. In some embodiments, the plurality of photovoltaic modules **110** is attached to the backsheet **134** by mechanical fasteners. In some embodiments, the fasteners may include nails, screws, rivets, or staples.

Still referring to FIG. **6**, in some embodiments, a first photovoltaic module **110***a* of the photovoltaic modules **110** is oriented such that it includes a negative terminal at the first end **112** and a positive terminal at the second end **114**. In some embodiments, a second photovoltaic module **110***b* of the photovoltaic modules **110** is oriented 180 degrees relative to the first photovoltaic module **110***a* such that includes a positive terminal at the second end **114** (now on the "left" side as shown in FIG. **6**) and a negative terminal at the first end **112** (now on the "right" side as shown in FIG. **6**). In some embodiments, a power electronics unit **150** is electrically connected to the first branch **127** (negative terminal) of the first photovoltaic module **110***a* and the second branch **129** (positive terminal) of the second photovoltaic module **110***b*. In some embodiments, the power electronics unit **150** is electrically connected to the first and second photovoltaic modules **110***a*, **110***b* in a manner similar to that as described above with respect to the power electronics units **50** to the photovoltaic shingles **10**. In some embodiments, the second branch **129** (positive terminal) of the first photovoltaic module **110***a* is electrically connected to the first branch **127** (negative terminal) of the second photovoltaic module **110***b*. In some embodiments, the second branch **129** (positive terminal) of the first photovoltaic module **110***a* is electrically connected to the first branch **127** (negative terminal) of the second photovoltaic module **110***b* by a jumper wire **136** (electrical jumper cable).

In some embodiments, a lower edge **115** of the photovoltaic module **110***b* overlays the section **131** of the first photovoltaic module **110***a*. In some embodiments, the lower edge **115** of the photovoltaic module **110***b* overlays the at least one bypass diode **130** of the first photovoltaic module **110***a* to reduce wasted space resulting from the section **131**.

In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes two of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes three of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes four of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes five of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes six of the photovoltaic modules **110**. In some embodiments, the plurality of photovoltaic modules **110** attached to the backsheet **134** includes more than six of the photovoltaic modules **110**.

FIG. **7** illustrates the photovoltaic shingle **100** having the photovoltaic module **110***b* with the string of the solar cells **120** thereof reversed. In some embodiments, the string of solar cells **120** is reversed during lamination of the photovoltaic module **110***b*. In some embodiments, the first photovoltaic module **110***a* overlaps the section **131** containing the at least one bypass diode **130**. In some embodiments, the first photovoltaic module **110***a* overlaps the section **131** thereof containing the at least one bypass diode **130** to reduce wasted space. FIG. **8** illustrates a plurality of the photovoltaic modules **110** installed on the roof deck with second photovoltaic module **110***b* having the reversed strings of solar cells **120**.

Referring to FIG. **9**, in some embodiments, a photovoltaic shingle **200** includes at least two photovoltaic modules **210***a*, **210***b* attached to backsheet **234**. In some embodiments, the backsheet **234** forms a headlap portion **216**. In some embodiments, each of the photovoltaic modules **210***a*, **210***b* includes a first end **212**, a second end **214** opposite the first end **212**, a first positive terminal **220***a* and a first negative terminal **220***b* proximate the first end **212**, and a second positive terminal **222***a* and a second negative terminal **222***b* proximate the second end **214**. In some embodiments, the second photovoltaic module **210***b* is vertically stacked and oriented 180 degrees relative to the first photovoltaic module **210***a*. Referring to FIG. **10**, in some embodiments, the photovoltaic shingle **200** includes a third photovoltaic module **210***c* attached to a backsheet **234**. In some embodiments, the third photovoltaic module **210***c* is vertically stacked and in the same orientation as the first photovoltaic module **210***a*. Referring to FIG. **11**, in some embodiments, the second photovoltaic module **210***b* is horizontally positioned adjacent to, and oriented 180 degrees relative to, the first photovoltaic module **210***a*.

What is claimed is:

1. A system, comprising:

a plurality of building integrated photovoltaic modules installed on a roof deck,

wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck,

wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module,

wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module,

wherein each of the plurality of building integrated photovoltaic modules includes

a plurality of solar cells arranged in at least one row, and

a power electronics unit,

wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells,

wherein the power electronics unit is electrically connected to the plurality of solar cells,

wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module,

wherein the power electronics unit includes a first terminal and a second terminal,

a first electrical connector electrically connected to the plurality of solar cells, and

a second electrical connector electrically connected to the plurality of solar cells,

wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module.

2. The system of claim **1**, wherein the power electronics unit is horizontally adjacent to the first end of the one of the at least one row of the plurality of solar cells.

3. The system of claim **1**, wherein the plurality of building integrated photovoltaic modules is a plurality of solar shingles.

4. The system of claim **1**, wherein each of the first electrical connector and the second electrical connector includes a first connector and a second connector, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module.

5. The system of claim 4, wherein the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the second electrical connector of the second building integrated photovoltaic module.

6. The system of claim 5, wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the first building integrated photovoltaic module.

7. The system of claim 6, wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is mechanically connected to the first connector of the second electrical connector of the second building integrated photovoltaic module.

8. The system of claim 7, wherein the second terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the first building integrated photovoltaic module by a first electrical cable.

9. The system of claim 8, wherein the second connector of the second electrical connector of the first building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the first building integrated photovoltaic module by a second electrical cable.

10. The system of claim 9, wherein the second terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second connector of the first electrical connector of the second building integrated photovoltaic module by a third electrical cable.

11. The system of claim 10, wherein the second connector of the second electrical connector of the second building integrated photovoltaic module is electrically connected to the first connector of the first electrical connector of the second building integrated photovoltaic module by a fourth electrical cable.

12. The system of claim 11, wherein the power electronics unit includes a third terminal and a fourth terminal, wherein the third terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the fourth terminal of the power electronics unit of the second building integrated photovoltaic module by a fifth electrical cable.

13. The system of claim 12, further comprising a third electrical connector installed on the roof deck, wherein each of the power electronics unit of the first building integrated photovoltaic module and the power electronics unit of the second building integrated photovoltaic module is electrically connected to the third electrical connector.

14. The system of claim 13, wherein the third electrical connector includes a first connector and a second connector, wherein the third terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the first connector of the third electrical connector by a sixth electrical cable.

15. The system of claim 14, wherein the fourth terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second connector of the third electrical connector by a seventh electrical cable.

16. The system of claim 1, wherein the power electronics unit includes a housing and power electronics within the housing.

17. The system of claim 16, wherein the power electronics includes an optimizer, a bypass diode, system monitoring electronic components, a rapid shutdown device, an electronic communication component, or combinations thereof.

18. The system of claim 1, wherein the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners.

19. The system of claim 1, wherein the plurality of building integrated photovoltaic modules includes a third building integrated photovoltaic module, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module.

20. The system of claim 19, wherein the plurality of building integrated photovoltaic modules includes a fourth building integrated photovoltaic module, wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module.

21. The system of claim 20, wherein the fourth building integrated photovoltaic module is vertically adjacent to the third building integrated photovoltaic module.

22. The system of claim 21, wherein the power electronics unit of the third building integrated photovoltaic module is electrically connected to the power electronics unit of the fourth building integrated photovoltaic module.

23. A system, comprising:
- a plurality of building integrated photovoltaic modules installed on a roof deck,
  - wherein the plurality of building integrated photovoltaic modules is arranged in an array on the roof deck,
  - wherein the plurality of building integrated photovoltaic modules includes a first building integrated photovoltaic module and a second building integrated photovoltaic module,
  - wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module,
    - wherein each of the plurality of building integrated photovoltaic modules includes
      - a plurality of solar cells arranged in at least one row,
      - a first electrical connector electrically connected to the plurality of solar cells, and
      - a second electrical connector electrically connected to the plurality of solar cells;
- a first power electronics unit,
  - wherein the first power electronics unit includes a first terminal and a second terminal,
  - wherein the first power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the first building integrated photovoltaic module,
  - wherein the first power electronics unit is electrically connected to the plurality of solar cells of the first building integrated photovoltaic module; and
- a second power electronics unit,
  - wherein the second power electronics unit includes a first terminal and a second terminal,
  - wherein the second power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells of the second building integrated photovoltaic module, and
  - wherein the first power electronics unit is electrically connected to the second power electronics unit, and

wherein the first terminal of the first power electronics unit is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module.

24. A method, comprising:

obtaining a plurality of building integrated photovoltaic modules,

wherein each of the plurality of building integrated photovoltaic modules includes

a plurality of solar cells arranged in at least one row, and

a power electronics unit,

wherein the power electronics unit is adjacent to a first end of one of the at least one row of the plurality of solar cells,

wherein the power electronics unit is electrically connected to the plurality of solar cells

wherein the power electronics unit includes a first terminal and a second terminal,

a first electrical connector electrically connected to the plurality of solar cells, and

a second electrical connector electrically connected to the plurality of solar cells; and

installing at least a first building integrated photovoltaic module and a second building integrated photovoltaic module on a roof deck,

wherein the first building integrated photovoltaic module is vertically adjacent to the second building integrated photovoltaic module, and

wherein the power electronics unit of the first building integrated photovoltaic module is electrically connected to the power electronics unit of the second building integrated photovoltaic module, and

wherein the first terminal of the power electronics unit of the first building integrated photovoltaic module is electrically connected to the second electrical connector of the first building integrated photovoltaic module, and wherein the first terminal of the power electronics unit of the second building integrated photovoltaic module is electrically connected to the second electrical connector of the second building integrated photovoltaic module.

25. The method of claim **24**, further comprising installing a third building integrated photovoltaic module and a fourth building integrated photovoltaic module on the roof deck, wherein the third building integrated photovoltaic module is horizontally adjacent to the first building integrated photovoltaic module, and wherein the fourth building integrated photovoltaic module is horizontally adjacent to the second building integrated photovoltaic module and vertically adjacent to the third building integrated photovoltaic module.

26. The method of claim **24**, wherein the plurality of building integrated photovoltaic modules is a plurality of solar shingles.

27. The method of claim **24**, wherein the plurality of building integrated photovoltaic modules is installed on the roof deck by a plurality of fasteners.

28. The system of claim **1**, wherein the at least one row includes a plurality of rows, wherein the plurality of rows includes a first row and a second row above the first row, wherein the first electrical connector is proximate to the first end of the first row and electrically connected to the first row of the plurality of solar cells, and the second electrical connector is proximate to a first end of the second row and electrically connected to the second row of the plurality of solar cells.

29. The system of claim **28**, wherein each of the plurality of rows includes a second end opposite the first end of the row, wherein each of the first building integrated photovoltaic module and the second building integrated photovoltaic module includes a first negative electrical terminal located at the first end of the first row thereof, a first positive electrical terminal located at the second end of the second row thereof, a second positive electrical terminal located at the first end of the second row thereof, and a second negative terminal located at the second end of the second row thereof.

30. The system of claim **23**, wherein each of the first power electronics unit and second power electronics unit includes an optimizer, a bypass diode, system monitoring electronic components, a rapid shutdown device, an electronic communication component, or combinations thereof.

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